Aquasense-X

By: Krish mahajan

ABSTRACT

Water scarcity is a growing global concern, exacerbated by inefficient usage and wastage. Many households, businesses, and industries lack real-time monitoring systems, leading to excessive consumption and unpredictable shortages. To address this issue, this project presents a smart water monitoring and management system powered by an **ESP8266 microcontroller**, ensuring efficient water utilization through real-time tracking and automated flow control.

The system integrates **input and output flow sensors** to measure water flow, allowing precise monitoring of supply and consumption. An **ultrasonic sensor** tracks storage tank levels, preventing dry runs and ensuring timely refilling. A **servo motor-operated valve** regulates distribution automatically, minimizing wastage and maintaining an optimal flow rate. Users can access real-time data through **a Blynk dashboard**, which provides insights into water availability, consumption patterns, and storage levels. Additionally, smart usage recommendations guide users on optimizing water consumption, ensuring sustainable usage.

For ease of control, the system offers manual and Auto Mode. In manual mode, users can adjust water flow settings via interactive dashboard buttons, while Auto Mode allows the system to autonomously regulate flow based on sensor data. The system ensures live monitoring of water input, output, and storage levels, providing accurate consumption insights. By integrating a servo-controlled valve, it automates water regulation to prevent excessive use. Furthermore, Wi-Fi auto-reconnect and a connectivity status indicator ensure seamless operation without disruptions. This versatile solution is applicable to homes, businesses, and industries, promoting sustainable water conservation through responsible usage.

3D Render Component Views

To visualize the system's operation, 3D render images highlight key components:

- **A: Main microcontroller unit (ESP8266)** – Controls system operations, processes sensor data, and communicates with the dashboard.
- **B: Input and output flow sensors** – Measure water flow at both entry and exit points, helping track usage and detect anomalies.
- **C: Servo motor-operated valve** – Regulates water flow automatically, optimizing distribution and minimizing wastage.
- **D: Ultrasonic sensor** – Monitors storage tank levels, ensuring timely refilling and preventing shortages.

By integrating these smart features, the system empowers users to take control of their water usage, minimize wastage, and contribute to sustainable water conservation. The combination of real-time monitoring, automated regulation, and intelligent recommendations makes this an efficient and responsible solution for modern water management.